

Dhanush S J

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SUMMARY

Software Engineering student with hands-on experience developing AI-powered and cloud-based applications using Python, Java, Flask, machine learning, and Large Language Models. Built end-to-end systems involving Retrieval-Augmented Generation, semantic search, document processing, REST APIs, databases, and Microsoft Azure services. Familiar with Scikit-Learn, PyTorch, vector search, data processing, and backend development, with a strong foundation in data structures, algorithms, problem solving, and software engineering. Interested in building reliable and scalable AI and software systems.

EDUCATION

Vellore Institute of Technology M.Tech Integrated Software Engineering	Expected 2027 CGPA: 8.13
NKM Higher Secondary School Class XII	2022 77%
Desia Matric High School Class X	2020 93.8%

LANGUAGES

English: Professional Working Proficiency	Tamil: Native Proficiency	Hindi: Conversational Proficiency
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TECHNICAL SKILLS

Programming: Python, Java, JavaScript, SQL
AI & LLMs: Retrieval-Augmented Generation (RAG), LangGraph, Hugging Face, TinyLlama, Google Gemini API
Machine Learning: Scikit-Learn, PyTorch (Basic), TensorFlow, Data Preprocessing, Feature Engineering, Model Training, Model Evaluation
Search & Data Processing: FAISS, Sentence Transformers, Semantic Search, Vector Search, Pandas, NumPy
Backend & APIs: Flask, Node.js, Express.js, REST APIs, SQLAlchemy
Databases & Cloud: MySQL, MongoDB, Firebase, Cloudinary, Microsoft Azure, Azure Blob Storage, Azure SQL Database
Developer Tools: Git, GitHub, Postman, Visual Studio Code

PROJECTS

- OrbitDesk AI Support Agent** | *Python, LangGraph, FAISS, Hugging Face, TinyLlama, Sentence Transformers*
- Created a local-first AI support system that answers user queries using internal knowledge-base documents and previously resolved support cases.
 - Designed a LangGraph workflow for query triage, semantic retrieval, response generation, verification, clarification, escalation, and out-of-scope handling.
 - Used FAISS and Sentence Transformers to retrieve relevant evidence and a locally hosted TinyLlama model to produce grounded responses.
 - Added verification and fallback logic to check evidence support, source references, and structured JSON output before returning a response.
- Azure Document Management System** | *Python, Flask, Azure Blob Storage, Azure SQL Database, REST APIs*
- Developed a cloud-based application for uploading, storing, retrieving, and managing documents through a web interface.
 - Integrated Azure Blob Storage for file storage and Azure SQL Database for maintaining document metadata and application records.
 - Implemented backend services and REST APIs using Python and Flask to handle document operations and database interactions.
 - Organized storage, database, and application components using a modular architecture to improve maintainability.
- Explainable AI for Clinical Diagnosis** | *Python, Scikit-Learn, XGBoost, SHAP, LIME, Flask*
- Developed a machine learning-based clinical decision support system for disease prediction using structured clinical data.
 - Performed data preprocessing, feature engineering, model training, and performance evaluation across multiple machine learning algorithms.
 - Applied SHAP and LIME to provide feature-level explanations and improve the interpretability of model predictions.
 - Created a Flask-based interface to present prediction results alongside human-readable explanations.
- AI Resume Analyzer and ATS Job Matcher** | *Python, Flask, Gemini, PyPDF2, REST APIs*
- Designed an AI-powered application that compares resumes with job descriptions and generates structured compatibility feedback.
 - Extracted information from PDF resumes to identify technical skills, compare job requirements, and highlight potential skill gaps.
 - Used prompt-based LLM processing to generate personalized recommendations and evaluation reports.
 - Exposed document processing and analysis functionality through Flask-based APIs.
- Real-Time IoT Monitoring and Surveillance System** | *Python, ESP32-CAM, Firebase, Gemini AI*
- Designed an event-driven monitoring system that captures images when motion is detected using an ESP32-CAM.
 - Processed captured events through a Python backend and integrated AI-generated descriptions for detected scenes.
 - Connected cloud storage and notification workflows to record events and deliver monitoring alerts.
 - Separated device interaction, backend processing, AI analysis, and event logging into modular system components.

CERTIFICATIONS & ACHIEVEMENTS

- Microsoft Certified: Azure Fundamentals (AZ-900)** — *Microsoft*; covering cloud concepts, Azure services, security, governance, pricing, and cloud architecture fundamentals.
- Generative AI Essentials: Using LLMs to Work with Data** — *IBM SkillsBuild*; covering LLM-based data analysis, structuring, and information extraction workflows.
- AI Fundamentals: Foundations for Understanding AI** — *IBM SkillsBuild*; covering AI fundamentals, machine learning concepts, and responsible AI practices.